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| Name : Harshad Khatale |
| PRN : 258209007 |
| Subject : Data Science & Analytics |
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Using a retail sales or energy consumption dataset, decompose the series into trend, seasonality, and residual components

Dataset - https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption

Description:

Time series forecasting is a statistical technique used to predict future values based on historical observations recorded over time. It is widely used in domains such as energy consumption, retail sales, stock market analysis, weather forecasting, and demand prediction. Time series data often exhibit patterns such as long-term trends, recurring seasonal effects, and random fluctuations. Identifying these components helps in understanding the behavior of the data and improves forecasting accuracy.

In this experiment, the **AEP Hourly Energy Consumption** dataset is used, which contains hourly electricity consumption (in megawatts) recorded over several years. The time series is decomposed into **Trend**, **Seasonality**, and **Residual** components to analyze its underlying structure. Forecasting models including a **Naive (Baseline)** model and the **Holt-Winters Exponential Smoothing** model are implemented. The forecasting performance is evaluated using **Mean Absolute Percentage Error (MAPE)** and **Root Mean Square Error (RMSE)**,

Theory

A **time series** is a sequence of observations collected at regular intervals over time. The primary objective of time series forecasting is to identify patterns in historical data and use them to predict future values.

1. Time Series Components

`a) Time trend`

The trend represents the long-term upward or downward movement in the data over time.

Example:

- Increasing electricity demand over several years.
- Continuous growth in retail sales.

b) Seasonality (S)

Seasonality refers to repetitive patterns occurring at fixed intervals.

Examples:

- Higher electricity consumption during daytime.
- Increased retail sales during festive seasons.

For hourly energy data, the seasonal period is typically **24 hours**.

c) Residual (R)

Residuals represent the random or irregular variations remaining after removing the trend and seasonal components.

These variations are caused by unexpected events or measurement noise

Conclusion

In this experiment, hourly energy consumption data was analyzed using time series forecasting techniques. The dataset was decomposed into **Trend**, **Seasonality**, and **Residual** components, providing a clear understanding of the underlying patterns in electricity demand. A **Naive forecasting model** was implemented as the baseline and compared with the **Holt-Winters Exponential Smoothing** model. Forecasting accuracy was evaluated using **MAPE** and **RMSE**, while actual and predicted values were visualized through comparison plots. The Holt-Winters model effectively captured both trend and seasonal variations, resulting in more accurate forecasts than the baseline model. This experiment demonstrates that incorporating trend and seasonality significantly improves forecasting performance for hourly energy consumption data.